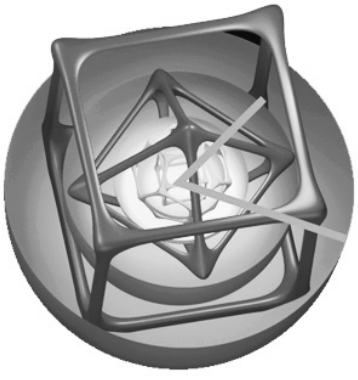
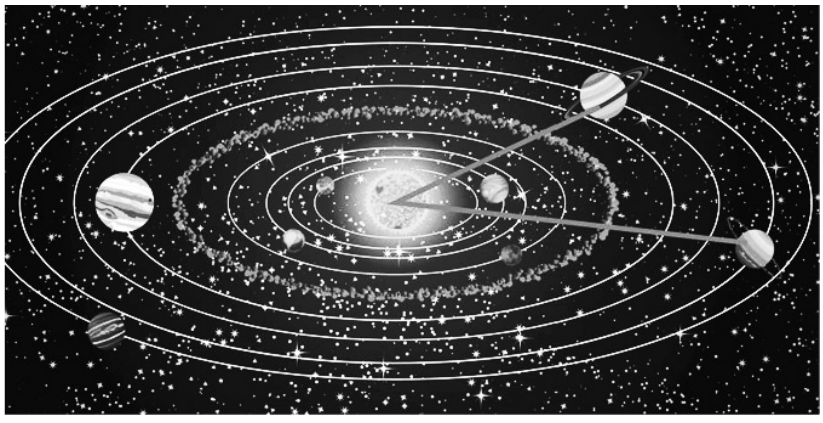


And he attempts to advance the audacious hypothesis that these are the actual shapes of the atoms of the five elementary substances that in antiquity were thought to form everything: earth, water, air, fire, and the quintessence of which the heavens are made. A beautiful idea. But completely mistaken. The error lies in seeking to understand the world in terms of things rather than events. It lies in ignoring change. The physics and astronomy that will work, from Ptolemy to Galileo, from Newton to Schrödinger, will be mathematical descriptions of precisely how things *change*, not of how they *are*. They will be about events, not things. The *shapes* of atoms will be eventually understood only with solutions to Schrödinger's equations describing how the electrons in atoms *move*. Events again, not things.

Centuries later, before reaching the momentous results obtained in his maturity, the young Kepler falls into the same error. He asks himself what determines the size of the orbits of planets and allows himself to be seduced by the same theorem that had bewitched Plato (it's a beautiful one, no doubt about that). Kepler assumes that the regular polyhedra determine the size of the orbits of the planets: if they are placed one inside the other with spheres between them, the radii of these spheres will be in the same proportion as the radii of the planets.





A nice idea, but completely wrongheaded. Once again, it lacks dynamics. When Kepler moves on, later in life, to tackle the question of how planets *move*, the gates of the heavens open.